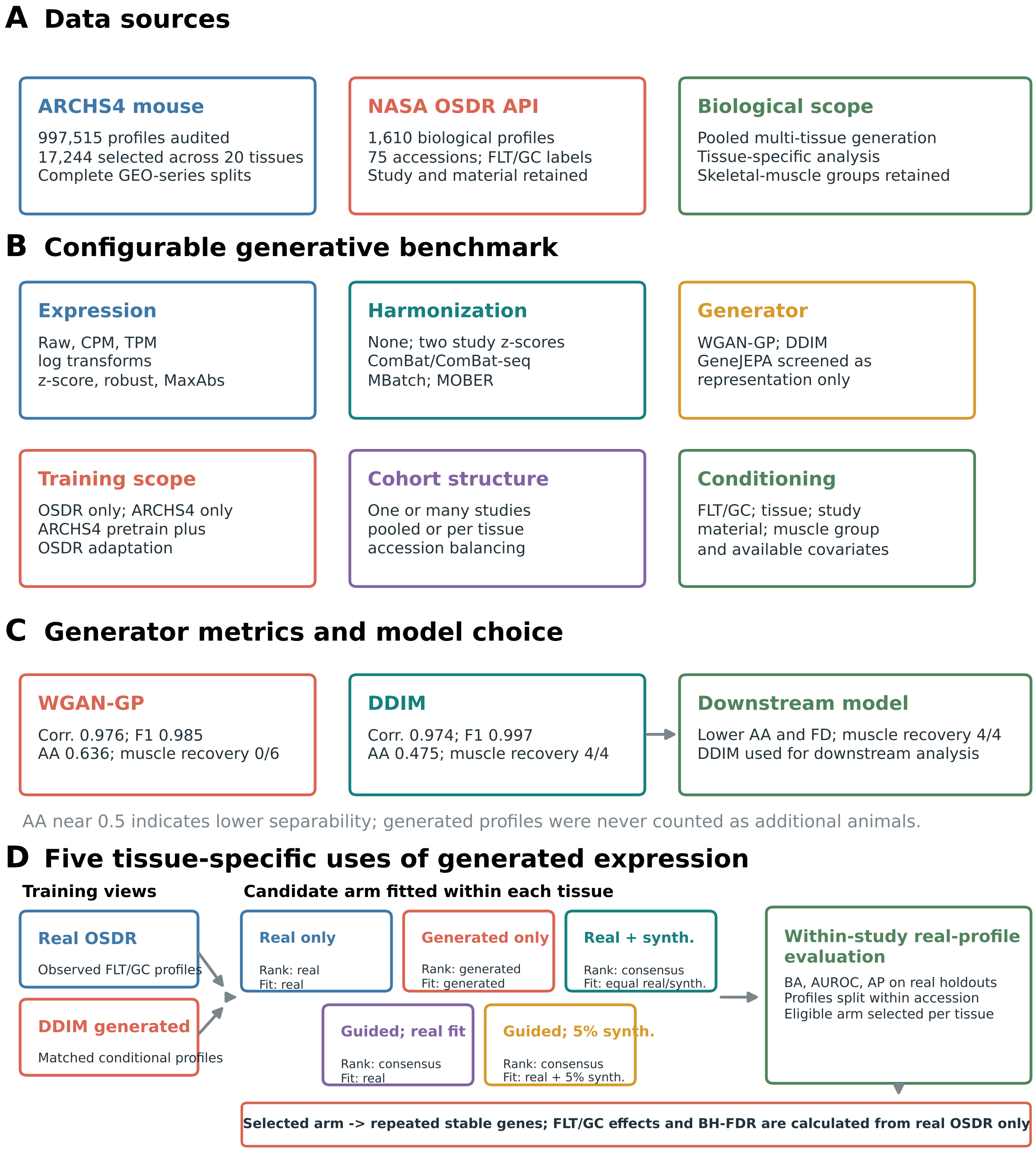
A Data sources
ARCHS4 mouse
997,515 profiles audited
17,244 selected across 20 tissues
Complete GEO-series splits
NASA OSDR API
1,610 biological profiles
75 accessions; FLT/GC labels
Study and material retained
Biological scope
Pooled multi-tissue generation
Tissue-specific analysis
Skeletal-muscle groups retained
B Configurable generative benchmark
Expression
Raw, CPM, TPM
log transforms
z-score, robust, MaxAbs
Harmonization
None; two study z-scores
ComBat/ComBat-seq
MBatch; MOBER
Generator
WGAN-GP; DDIM
GeneJEPA screened as
representation only
Training scope
OSDR only; ARCHS4 only
ARCHS4 pretrain plus
OSDR adaptation
Cohort structure
One or many studies
pooled or per tissue
accession balancing
Conditioning
FLT/GC; tissue; study
material; muscle group
and available covariates
C Generator metrics and model choice
WGAN-GP
Corr. 0.976; F1 0.985
AA 0.636; muscle recovery 0/6
DDIM
Corr. 0.974; F1 0.997
AA 0.475; muscle recovery 4/4
Downstream model
Lower AA and FD; muscle recovery 4/4
DDIM used for downstream analysis
AA near 0.5 indicates lower separability; generated profiles were never counted as additional animals.
D Five tissue-specific uses of generated expression
Training views
Candidate arm fitted within each tissue
Real OSDR
Observed FLT/GC profiles
DDIM generated
Matched conditional profiles
Real only
Rank: real
Fit: real
Generated only
Rank: generated
Fit: generated
Real + synth.
Rank: consensus
Fit: equal real/synth.
Guided; real fit
Rank: consensus
Fit: real
Guided; 5% synth.
Rank: consensus
Fit: real + 5% synth.
Within-study real-profile evaluation
BA, AUROC, AP on real holdouts
Profiles split within accession
Eligible arm selected per tissue
Selected arm -> repeated stable genes; FLT/GC effects and BH-FDR are calculated from real OSDR only